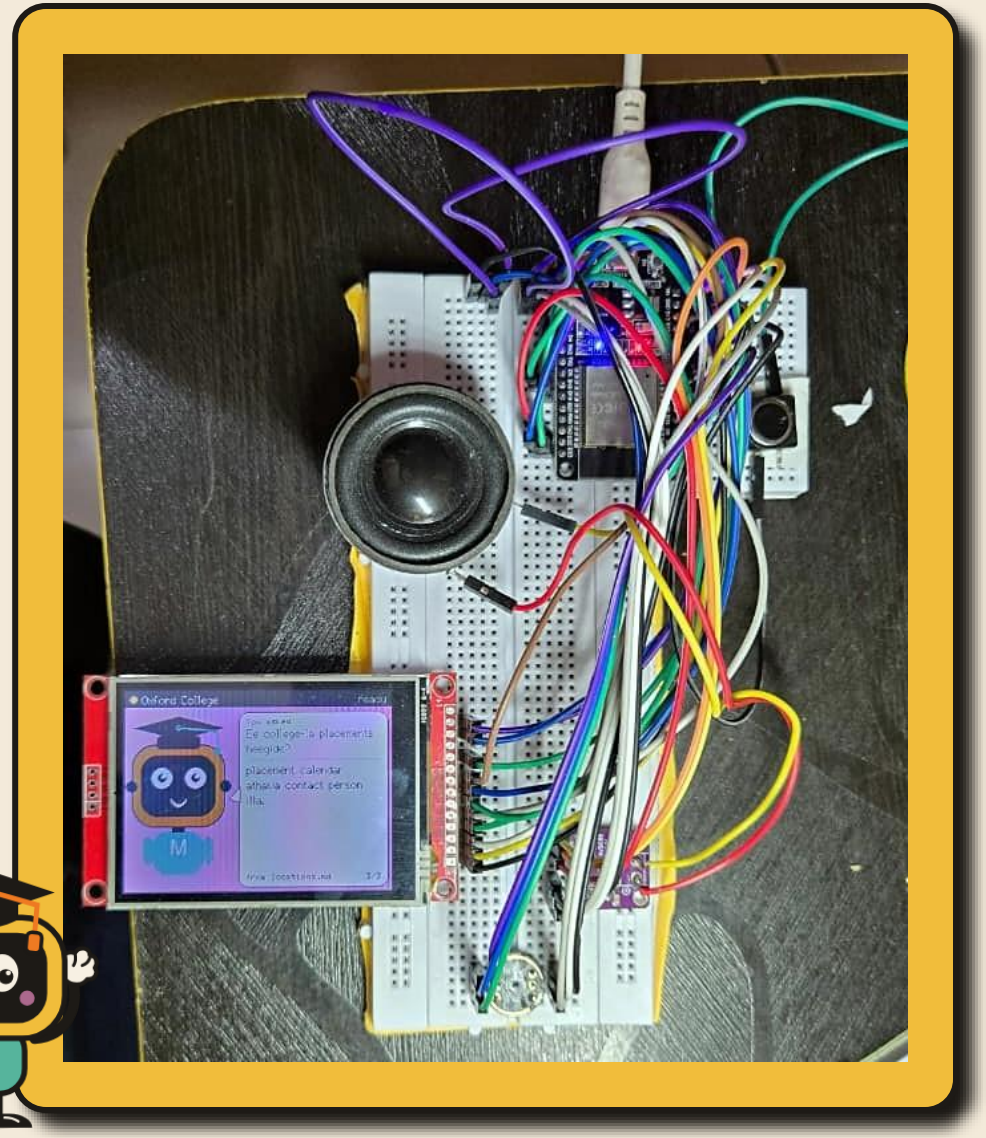
FIRSTHACK 2026 · SUBMISSION

CAMPUS MITHRA

A voice kiosk, app, website and admin dashboard that answer visitors' and students' campus questions in English, Kannada or Hindi, and walk them to the right building.

TEAM	Team Spy
TRACK	Campus and Student Life
COLLEGE	The Oxford College of Engineering, Bengaluru
MEMBERS	Yadava HC · Pavan Kumar H · Lalitha K · Bhavani Patil



first-hack-team-spy-sigma.vercel.app

02 THE PROBLEM

IMPACT & USEFULNESS · 20%

Admission day: families on a campus they have never seen, with questions nobody at the gate can answer in their language.

1 Lost on campus

Admission office, accounts, hostel, departments. Families walk around for twenty minutes looking for the right building.

2 Same questions, every year

Placements? CSE fee? Is there a hostel? A bus to my area? The college has every answer, in documents nobody at the gate can see.

3 Not in their language

Many parents are more at home in Kannada or Hindi. Many students speak Kanglish or Hinglish. An English notice board does not help.

WHO: first-time visitors on admission day — students and their parents — plus current students who need the hostel, a lab or today's notice.

HOW WE KNOW: *Add one real line from a parent or student you asked (who, and what they said).*

03 OUR SOLUTION

INNOVATION & ORIGINALITY · 30%

Campus Mithra lets a visitor ask the gate kiosk out loud, in their own language, and get a checked answer from the college's own records — with a QR code that walks them there on their phone.

WHAT MAKES IT DIFFERENT

- Real hardware at the gate: an ESP32 kiosk with a mic, speaker and a talking Mithra face — no phone or app needed to ask.
- Code-mixed speech: asked in Kanglish or Hinglish, it answers in Kanglish or Hinglish; plain Kannada or Hindi gets plain replies.
- Never makes things up: a grounding check rejects any sentence or number not in the college's records.
- One live source: kiosk, website, Android app and admin dashboard share the same campus data and map.

WHAT IT DOES NOT DO

- Answer anything outside the college — it says "I don't know about that."
- Run alone: the kiosk needs a PC beside it over USB.
- Show Kannada or Hindi script on the kiosk screen — it shows Latin letters and speaks the language.
- iOS: the app is Android only for now.

04 HOW IT WORKS

PRESENTATION & CLARITY · 20%

1 ASK

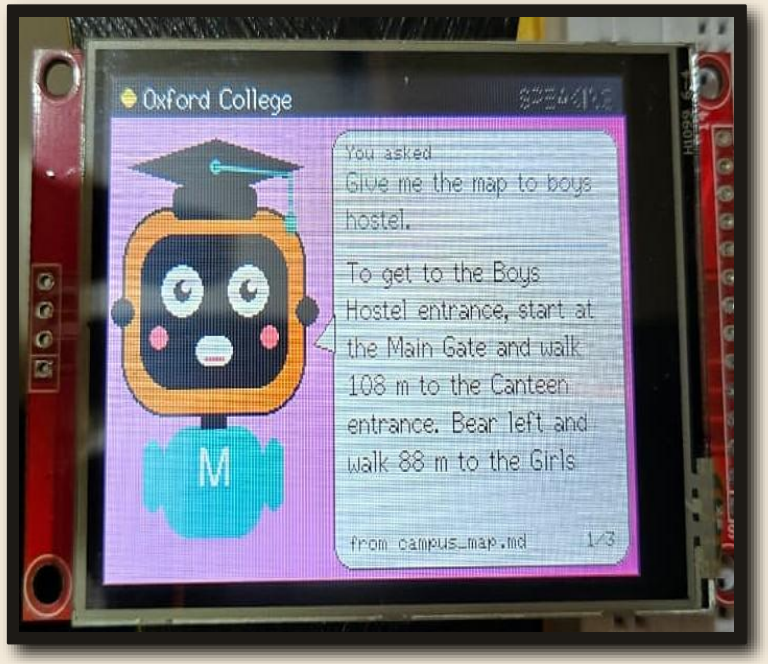
Hold the kiosk button and ask out loud: "Boys hostel-ge hoguva map kodu."

2 GET THE ANSWER

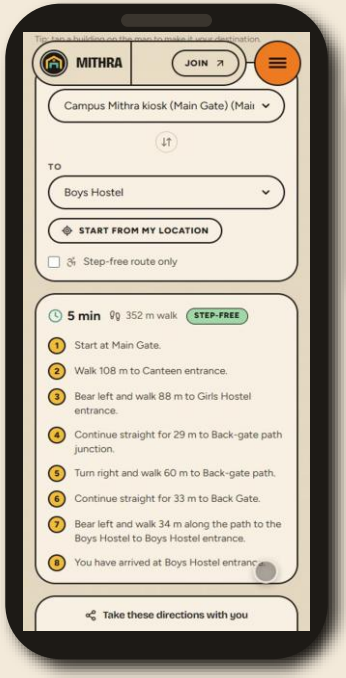
Mithra says and shows the answer in the same language, names the source, and shows a QR code.

3 WALK THERE

Scan it: the route opens on the phone with live GPS along the real campus paths.







5 min
352 m walk

05 WEBSITE AND APP, LIVE

TECHNICAL EXECUTION · 30%





LIVE CAMPUS MAP

Real campus paths on the satellite view, shortest and step-free routes, and a QR to take them to the phone.

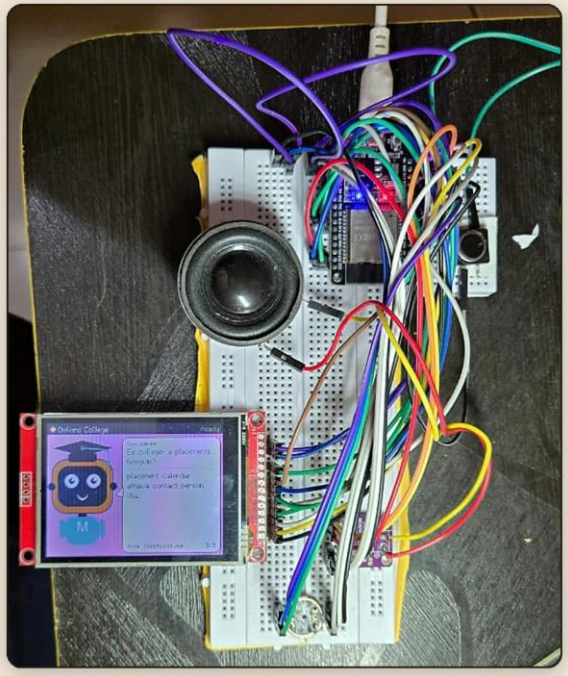
ANDROID APP · NOTICES · EVENTS

06 THE KIOSK HARDWARE

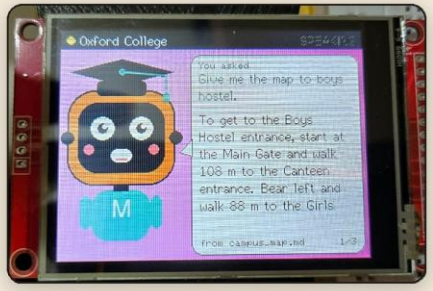
TECHNICAL EXECUTION · 30%

Campus Mithra kiosk

ESP32 voice kiosk: ask in English, Kannada or Hindi, get the answer and a walking-route QR code



The build: ESP32, 2.8" TFT, INMP441 mic, MAX98357A amp, speaker, talk button



Answers on screen and out loud, with the source



Scan for live walking directions on your phone

- **ESP32 DevKit** runs the firmware (C++, PlatformIO)
- **2.8" ILI9341 TFT** Mithra's face, answers and QR codes
- **INMP441 I2S mic** records the question at 16 kHz
- **MAX98357A amp + speaker** speaks the answer, lip-synced
- **Push button** push-to-talk on GPIO27

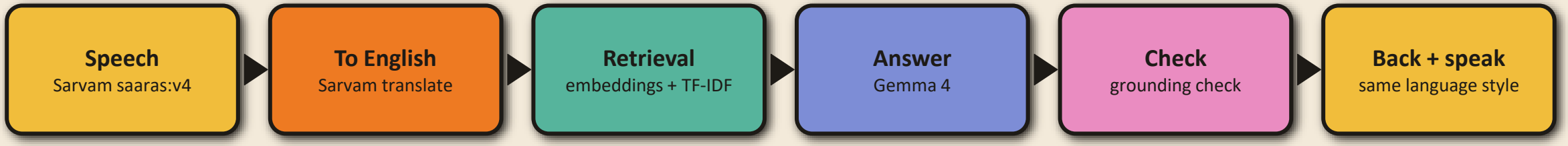
₹2–3k
kiosk electronics

USB 921600
baud to the PC — no Wi-Fi keys on the device

Less than one temporary helpdesk person for an admission season.

07 WHAT WE BUILT IT WITH

TECHNICAL EXECUTION · 30%



LANGUAGES & FRAMEWORKS

- Kiosk: C++ on ESP32 (PlatformIO, TFT_eSPI, QRCode)
- Server: Python, FastAPI, NumPy
- Web + admin: Next.js 16, React, TypeScript, Tailwind, Google Maps
- App: Flutter (Android)
- Backend: Firebase Auth, Firestore, Storage, FCM

WHAT WAS HARDEST

- Answering in the style asked: Kanglish in, Kanglish out — translate to English for search and checking, then back with mayura:v1 for code-mixed or sarvam-translate for pure Kannada/Hindi.
- Streaming speech to the ESP32 over USB serial without stutter: playback paced by credits from the kiosk.

WHAT WE USED FROM OTHERS

- Sarvam AI (speech, translation), Gemma 4 via Google AI Studio, edge-tts voices
- Remotion for the animated characters
- Open-source libraries listed in the repo
- AI coding assistants helped write and review code

08 SEE IT RUNNING

TECHNICAL EXECUTION · 30%

CODE REPOSITORY

Public, with full commit history.

github.com/Pavankumarh0/first-hack-teamSPY

DEMO VIDEO

Real kiosk, website, app and dashboard.

Paste the video link

LIVE LINK

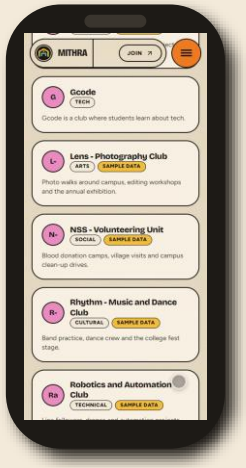
Website, admin, and the Android APK at /app.

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09 WHAT WORKS, WHAT'S NEXT

WORKING TODAY

- Kiosk answers spoken questions in English, Kannada, Hindi, Kanglish and Hinglish — on screen and out loud, with the source named
- Walking-route QR codes that open live GPS directions on the phone
- Live website and Android app: map, notices, events, clubs, issue reports, community, Campus IQ
- Admin dashboard: map editor, content, QR codes, kiosk device keys, moderation
- Push notifications when a notice is published

NOT BUILT YET

- Stand-alone kiosk on Wi-Fi, no PC — one box at every gate
- Kannada and Hindi script on the kiosk screen
- An enclosure for the breadboard build
- iOS app
- Admissions insights for the college: what visitors asked, in which language

5 ways to ask: English, Kannada, Hindi, Kanglish, Hinglish

12 college documents in the knowledge base

1 live data source for kiosk, web, app and admin

0 answers allowed that are not in the records

10 HOW A COLLEGE ADOPTS IT

IMPACT & USEFULNESS · 20%

WEEK 1

Map the campus: paths, buildings, rooms

WEEK 2

Load fees, placements, hostel, transport, events

WEEK 3

Install kiosks, print QR codes for every door

ADMISSION DAY

Visitors ask, scan and walk

BUSINESS MODEL (PROPOSED)

Software subscription: ₹50,000 – ₹1,50,000 a year per campus, by size

Kiosk hardware kit: one-time; sold or rented for the admission season

Setup service: we map the campus and load the content

Admissions insights: premium add-on

WHERE WE GROW



Anywhere people get lost and ask the same questions: hospitals, government offices, malls, exhibitions and events.

11 THE TEAM

YH

Yadava HC

Year & branch

What they built

PK

Pavan Kumar H

Year & branch

What they built

LK

Lalitha K

Year & branch

What they built

BP

Bhavani Patil

Year & branch

What they built

TEAM SPY · THE OXFORD COLLEGE OF ENGINEERING, BENGALURU

CAMPUS MITHRA · TEAM SPY

THANK YOU

Your friend on campus, in your language.

Press a button, ask in your own language, get a true answer from the college itself, and walk straight there.



first-hack-team-spy-sigma.vercel.app

github.com/Pavankumarh0/first-hack-teamSPY

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The Oxford College of Engineering, Bengaluru